

MATH FOR AI • EPISODE 0.3

Functions as Machines

Input goes in. Output comes out.

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Last time's challenge: the answer

$$\begin{aligned}\sum_{i=1}^5 x_i y_i &= 1 \cdot 49 + 2 \cdot 56 + 3 \cdot 65 + 4 \cdot 71 + 5 \cdot 82 \\ &= 49 + 112 + 195 + 284 + 410 = 1050\end{aligned}$$

For each student: hours **times** score. Then add them all up.

Every AI model is a function

$$\hat{y} = f(x)$$

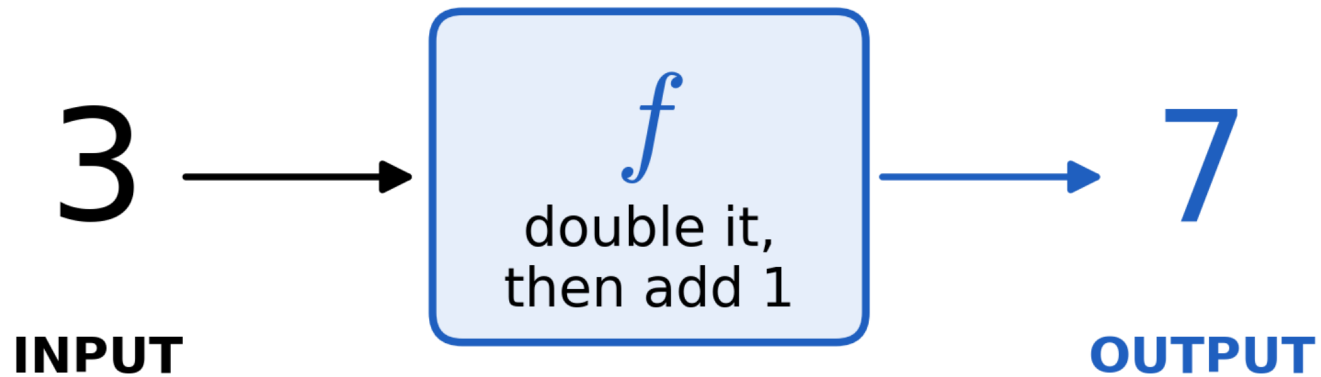
A photo goes in → **“cat”** comes out.

A sentence goes in → **a score for every possible next word** comes out.

Hours studied go in → **a predicted score** comes out.

Different inputs, different outputs. Same big idea: **a function**.

A function is a machine



$$f(3) = 2 \cdot 3 + 1 = 7$$

THREE PARTS

Input: what you put in

Rule: what the machine does to it

Output: what comes out

Reading $f(x)$ out loud

SAY IT

$f(x)$

“f of x”

WHAT EACH PIECE MEANS

f is the machine’s **name**

() right after the name is the **input chute**

x is **what goes in**

f(x) is **what comes out**

Heads up: $f(x)$ is **not** “f times x.” The parentheses mean “feed x into f.”

Plugging in: swap x for a number

THE RULE

$$f(x) = 2x + 1$$

INPUT 3

$$f(3) = 2 \cdot 3 + 1 = 7$$

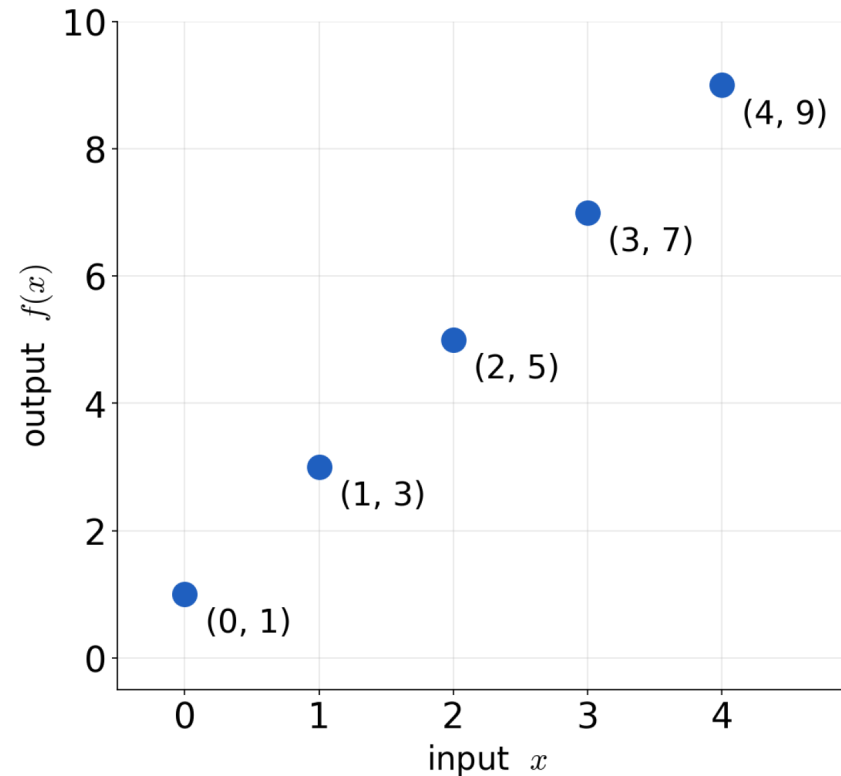
INPUT 10

$$f(10) = 2 \cdot 10 + 1 = 21$$

Every x in the rule gets replaced by the input. Then do the arithmetic.

A table of inputs and outputs

x	$f(x)$
0	1
1	3
2	5
3	7
4	9



EACH ROW IS A DOT

Input across, output up.

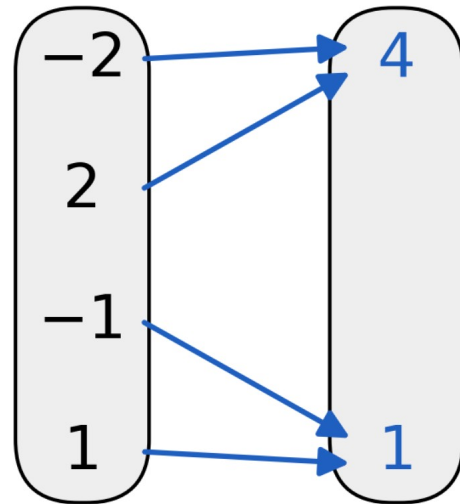
The dots line up in a straight line.

Why a line? That's Episode 0.4.

The one rule: one input, one output

A function ✓

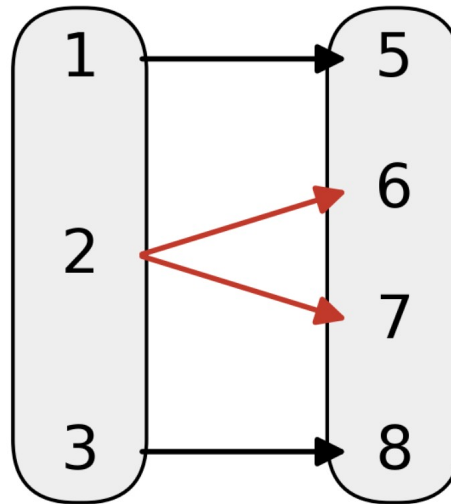
rule: multiply it by itself



every input: exactly one arrow out

Not a function ✗

one input, two outputs



the input 2 has TWO arrows out

A FUNCTION NEVER GUESSES

Same input in, same output out.
Every time.

Two inputs **can** share an output (-2 and 2 both give 4). One input can't have two.

A function in Python

MATH

$$f(x) = 2x + 1$$

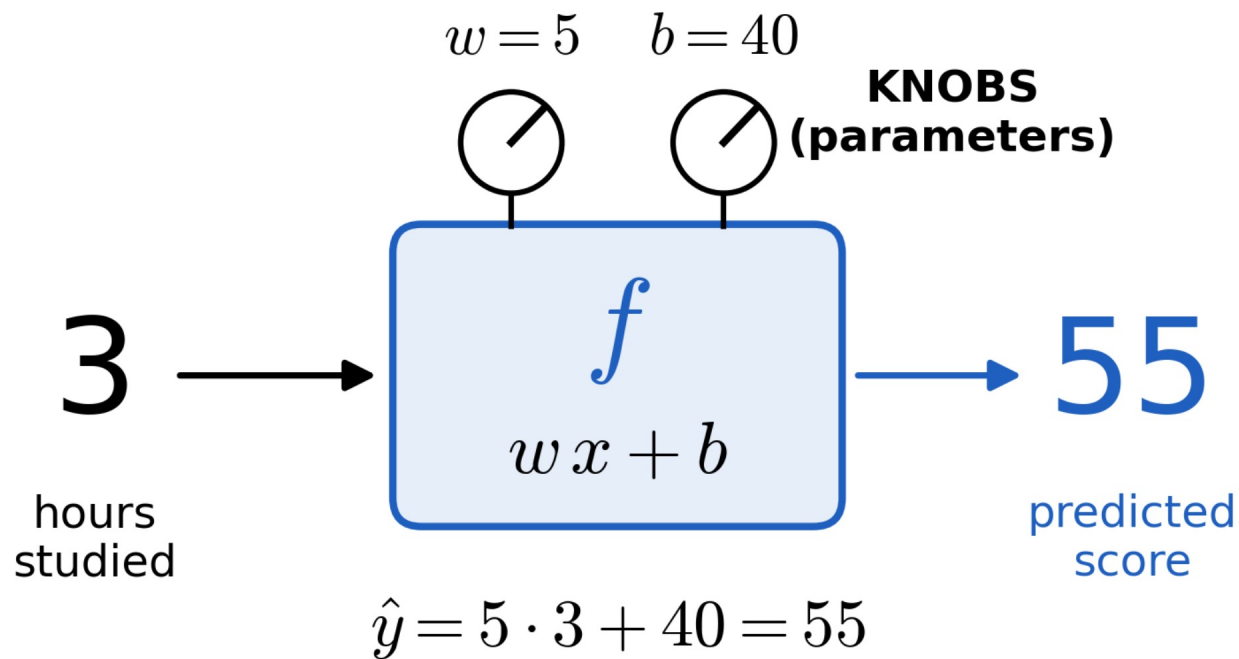
PYTHON

```
def f(x):                # build a machine name
    d f
    return 2 * x + 1 # the output chute

print(f(3))              # 7
print(f(10))             # 21
```

def builds the machine. **return** is the output chute. **f(3)** means the same thing in both languages.

Our AI model is a machine with knobs

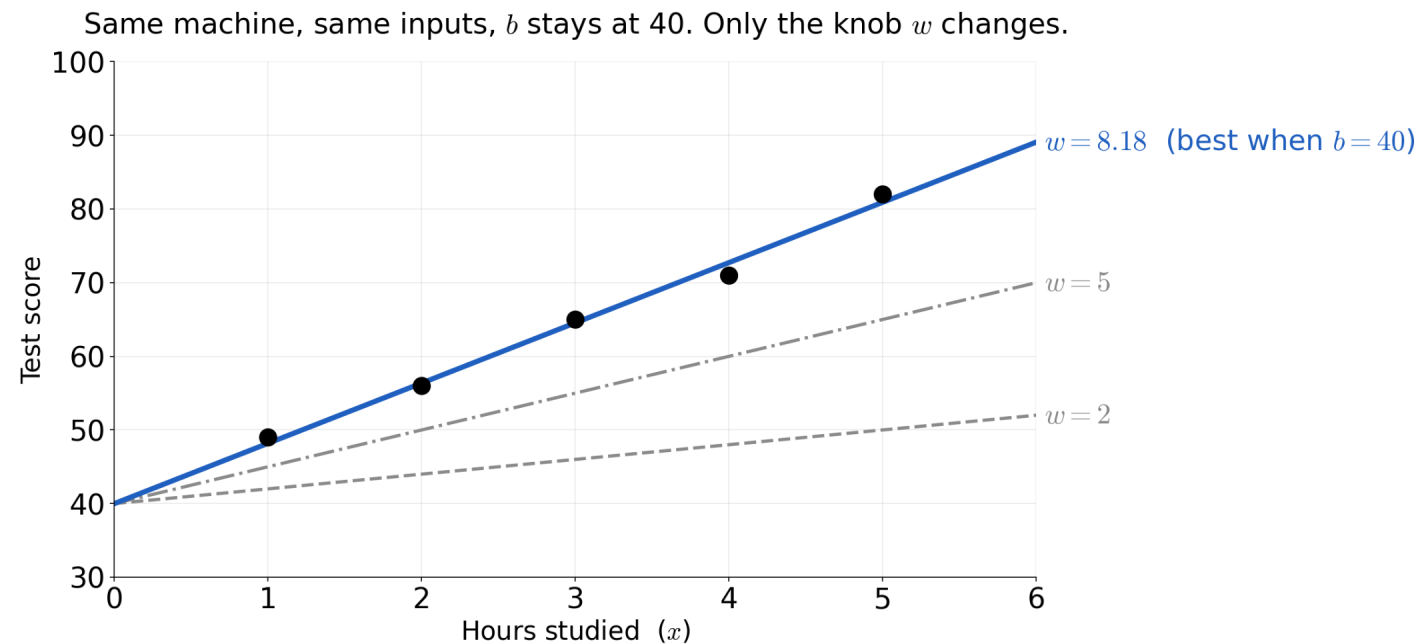


$$\hat{y} = f(x) = w x + b$$

$\mathbf{x}$ is the input: it changes for every student.

$\mathbf{w}$ and $\mathbf{b}$ are **knobs** (parameters): set on the machine.

Training = turning the knobs



SAME INPUTS, NEW KNOB

Turn **w** and every output changes.

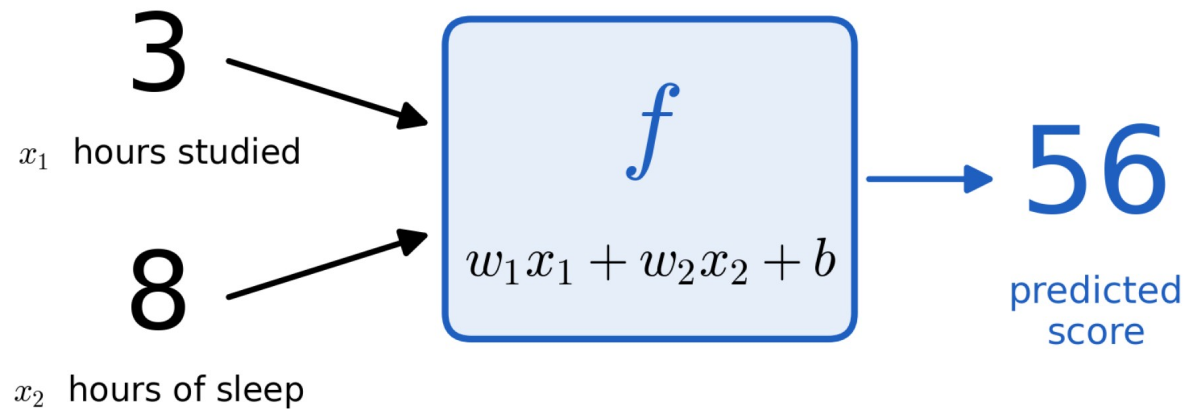
Learning = turning the knob until the outputs get as close to the real scores as they can.

Here **b** stays at 40 the whole time.

Machines can have more than one input

$$f(x_1, x_2) = w_1x_1 + w_2x_2 + b$$

(made-up knobs: $w_1 = 4$, $w_2 = 3$, $b = 20$)



$$f(3, 8) = 4 \cdot 3 + 3 \cdot 8 + 20 = 56$$

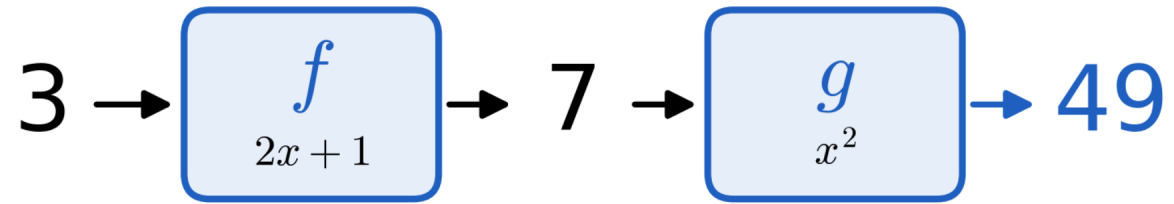
A comma separates the inputs.

Each input gets **its own knob**.

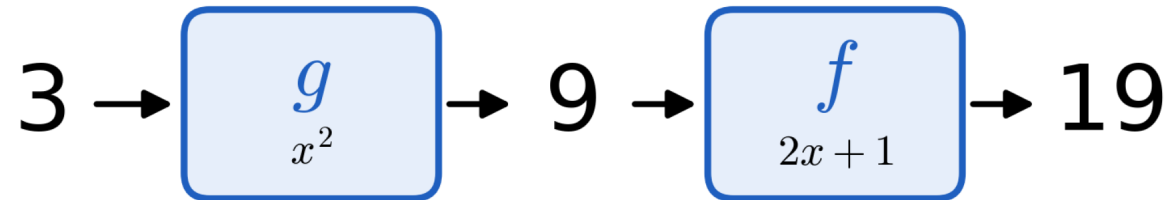
Episode 0.2's weighted sum is this machine with 2 inputs.

Machines can be chained

$$g(f(3)) = g(7) = 49$$



$$f(g(3)) = f(9) = 19$$



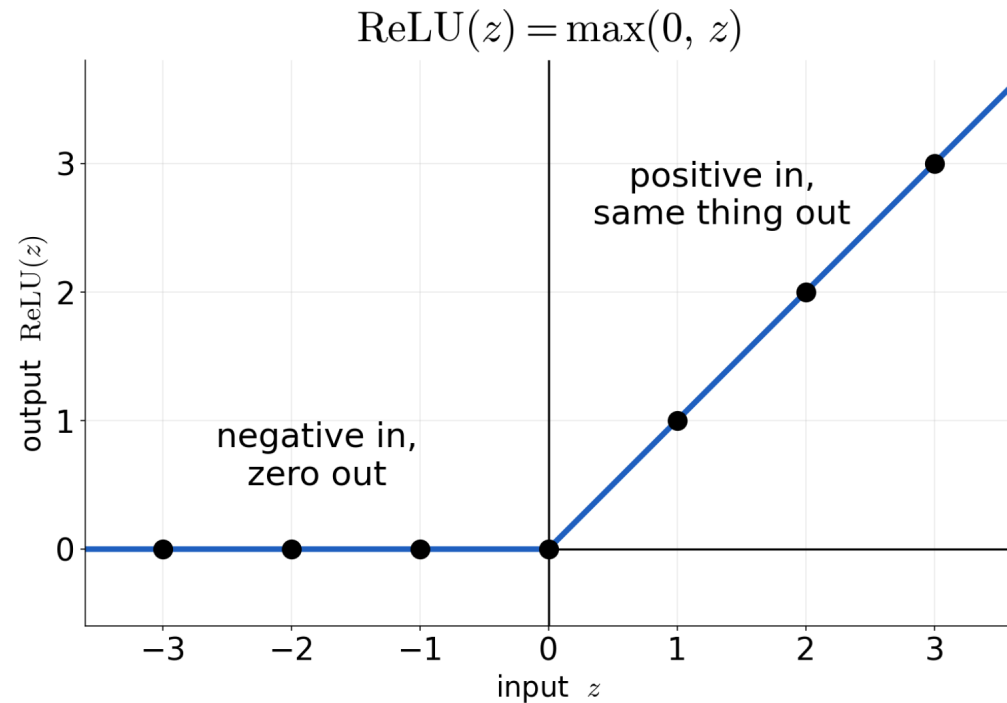
$$g(x) = x^2$$

The output of one machine becomes the input of the next.

Work from the inside out.

Order matters: $49 \neq 19$.

ReLU: a real machine inside neural networks



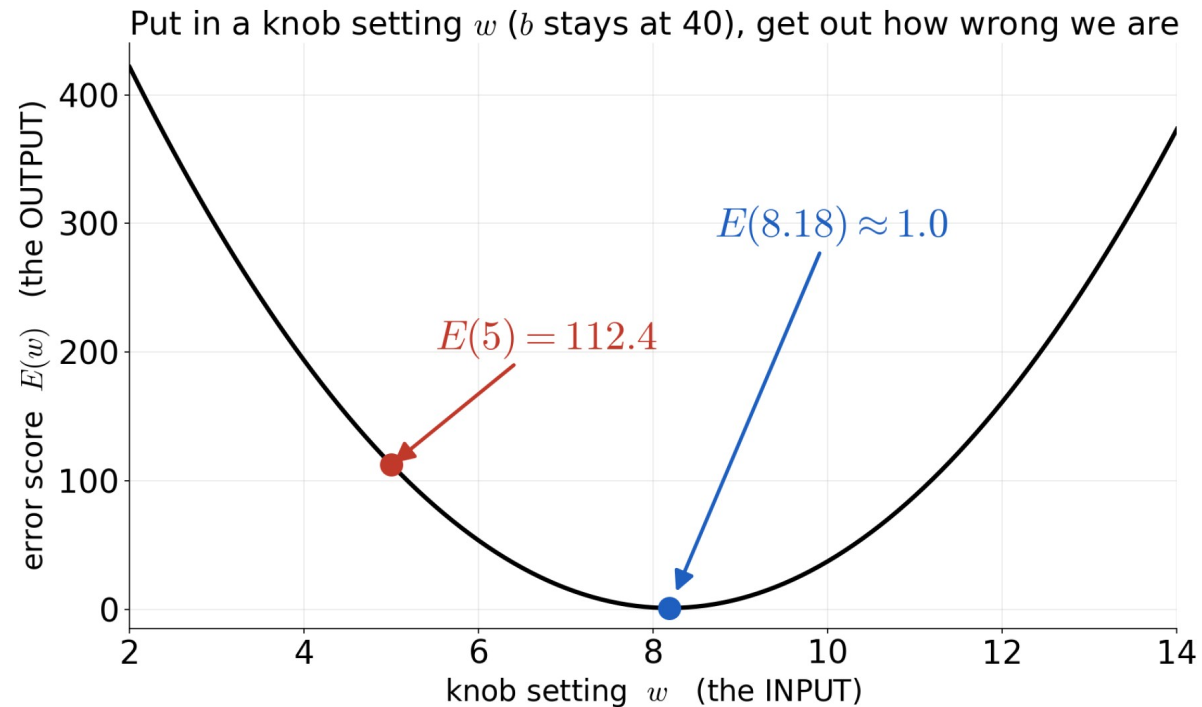
$$\text{ReLU}(z) = \max(0, z)$$

max means “pick the bigger one.”

Zero or negative in $\rightarrow$ **0** out.

Positive in $\rightarrow$ **same** out.

The error is a function too



$$E(w) = \frac{1}{n} \sum_{i=1}^n (y_i - (w x_i + b))^2$$

Input: a knob setting $\mathbf{w}$.

Output: how wrong we are.

w is still the model's knob. For the error machine, it's the input. The students' data stays fixed, and b stays at 40.

Training = find the input that gives the **smallest** output.

Your turn: run the machines

1 $f(x) = 3x - 2.$ $f(4) = ?$

2 $f(x) = 2x + 1,$ $g(x) = x^2.$ $g(f(2)) = ?$

3 $\text{ReLU}(-7) + \text{ReLU}(7) = ?$

Pause the video. Say each one out loud, then work it out.

For #2, start with the inside parentheses.

Answers

$$1 \quad f(4) = 3 \cdot 4 - 2 = 10$$

$$2 \quad g(f(2)) = g(5) = 25$$

$$3 \quad \text{ReLU}(-7) + \text{ReLU}(7) = 0 + 7 = 7$$

Got all three? You can now run any function by hand.

Check them in Python: every answer is one line.

Real AI formulas you can now read

$$\hat{y} = f(x)$$

A model is a machine: input in, prediction out.

$$a = \text{ReLU} \left[\sum_{j=1}^d w_j x_j + b \right]$$

One neuron: a weighted sum, chained into ReLU.

$$\hat{y} = f_3(f_2(f_1(x)))$$

A deep network: machines chained, layer by layer.

THE EQUATION OF THE DAY

$$\hat{y} = f(x)$$

Every AI model is a function: an input goes in, and a prediction comes out.

NEXT TIME · EPISODE 0.4

Graphing functions

Lines, parabolas, and what their shapes tell you.

YOUR CHALLENGE · answer in Episode 0.4

$$g(x) = x^2 \quad x = -2, -1, 0, 1, 2$$

Make the table. Plot the five dots on paper. What shape do you see?